

Spatial Dataflow at Wafer Scale: SpMV on the Cerebras WSE

Maya Taylor, Luke Olson
University of Illinois at Urbana-Champaign

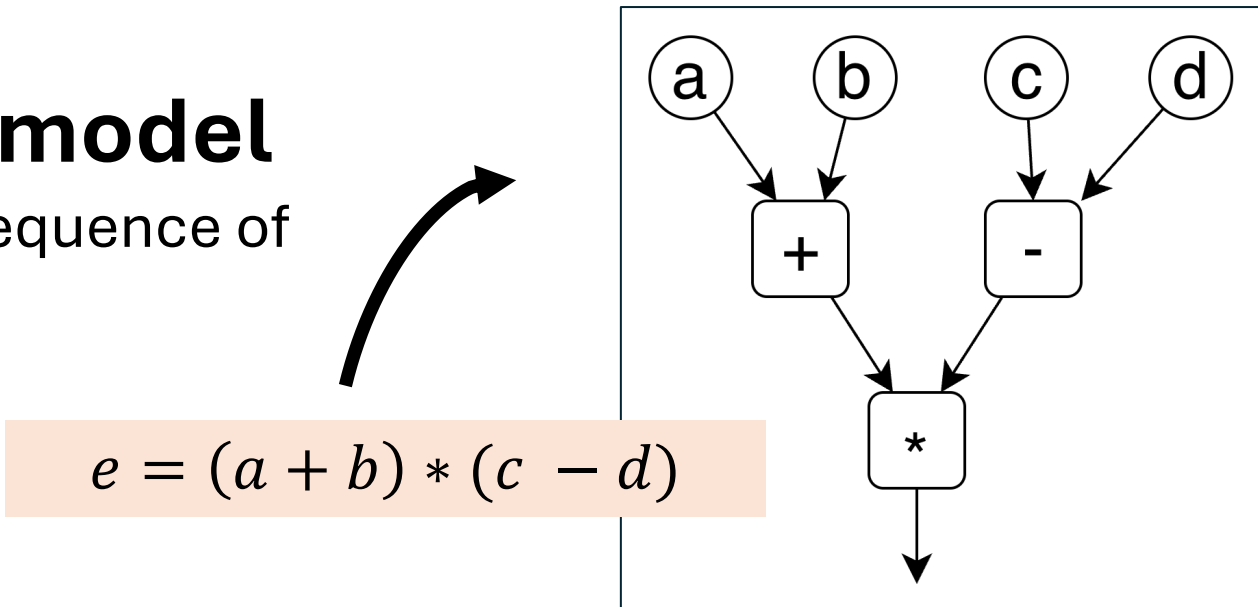
Second-Year Fellow
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Dataflow has many meanings in the world of computer science...

Dataflow as a programming model

A program is represented as a graph, not a sequence of instructions as in the von Neumann model

- nodes are operators
- edges are data dependencies
- no program counter dictating order
- a node executes as soon as all its input tokens are available
- independent subgraphs can fire concurrently creating implicit parallelism



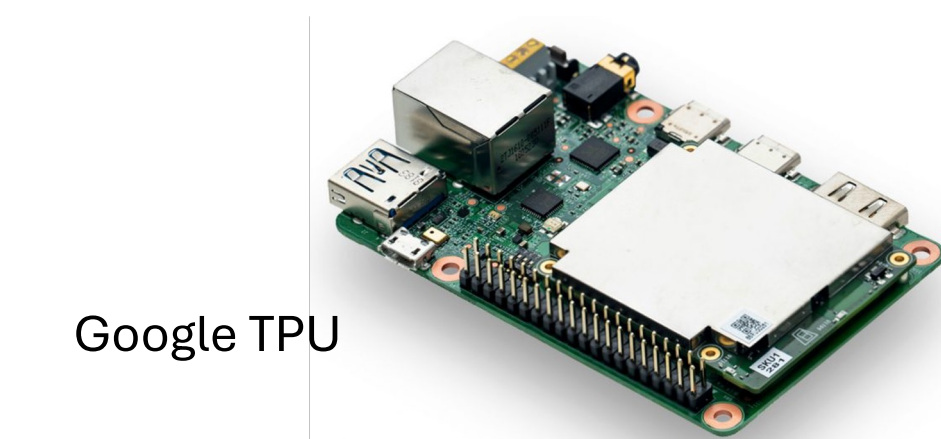
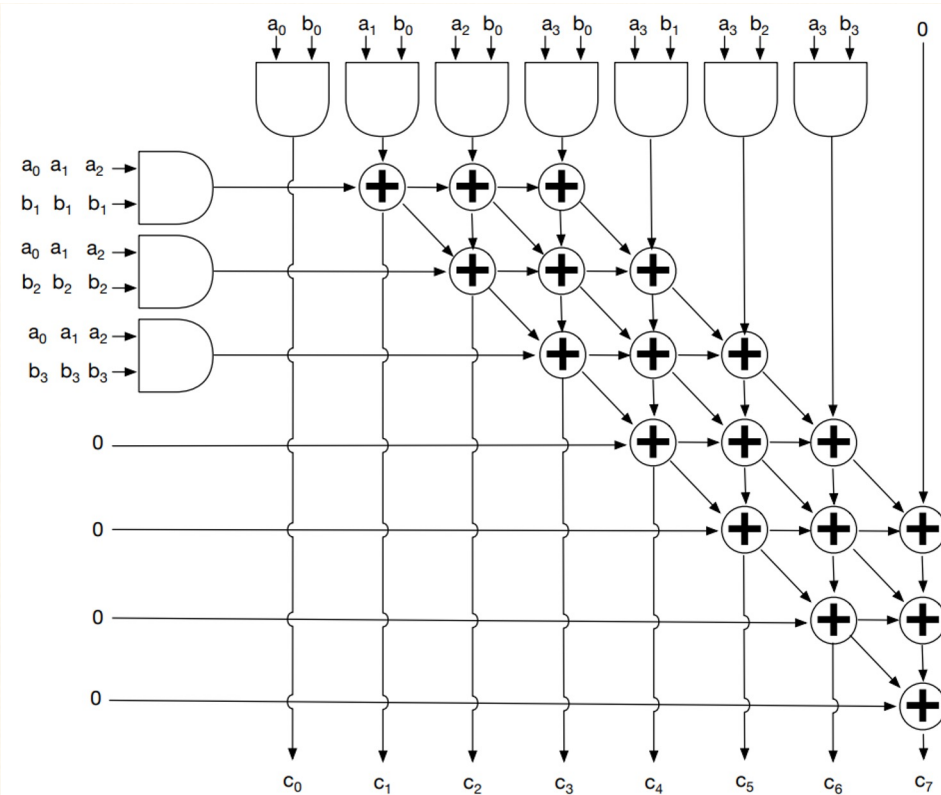
Early dataflow machines

Manchester Dataflow Machine (1981)

- first working dynamic/tagged-token prototype

MIT TTDA (mid-1980s)

- scaled the tagged-token idea to a multiprocessor
- each token carries a tag identifying its loop iteration or call frame



Google TPU

Systolic arrays

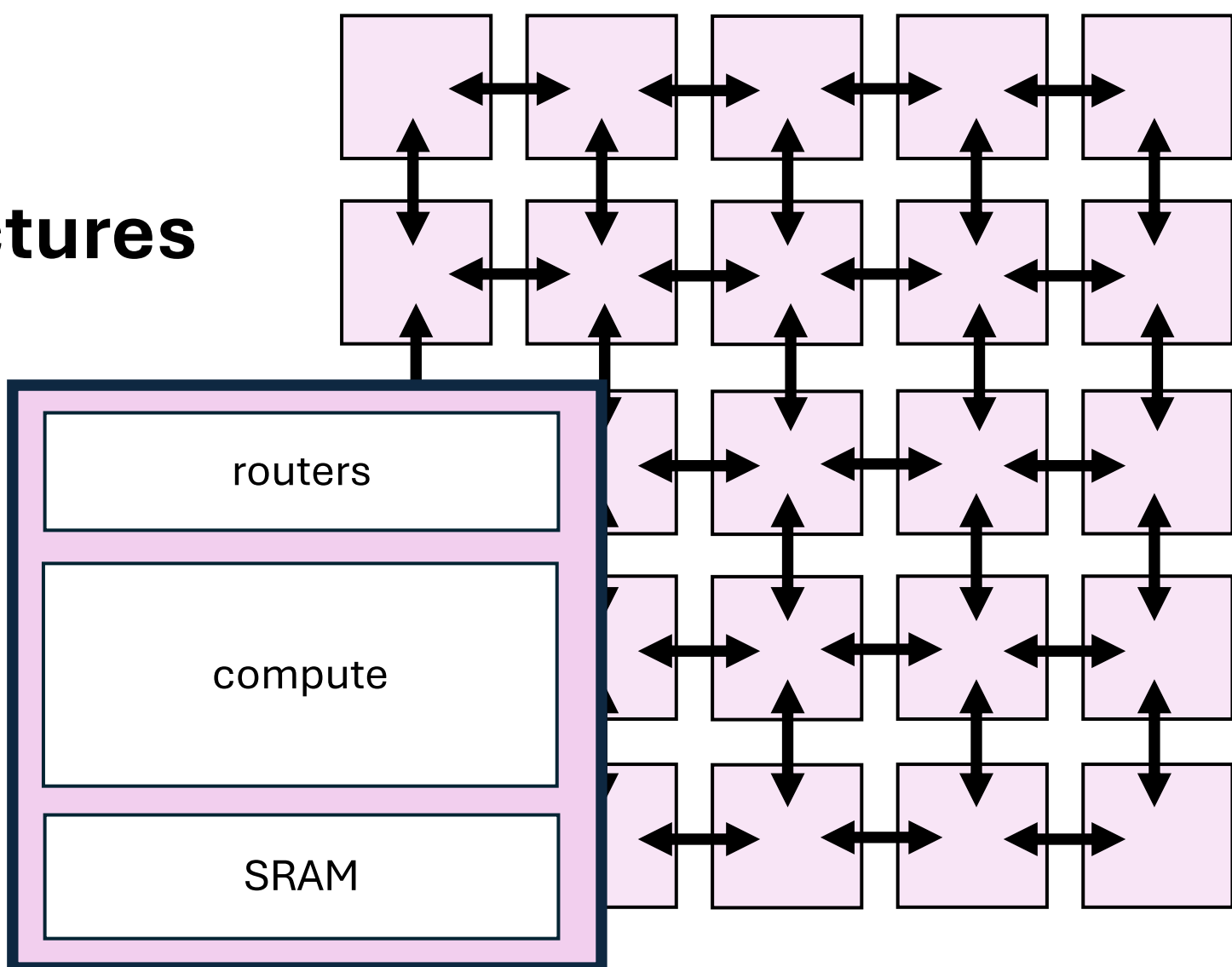
Network of tightly coupled compute nodes

- rhythmically pass data in lockstep
- nodes usually have identical, limited compute capabilities
- the Google TPU is designed around a systolic array for matrix multiplication

Spatial dataflow architectures

More general than a systolic array

- 2D mesh of compute nodes
- more complex compute
- can express more general computation
- not globally clocked
- designed for AI
- initial success for numerical workloads

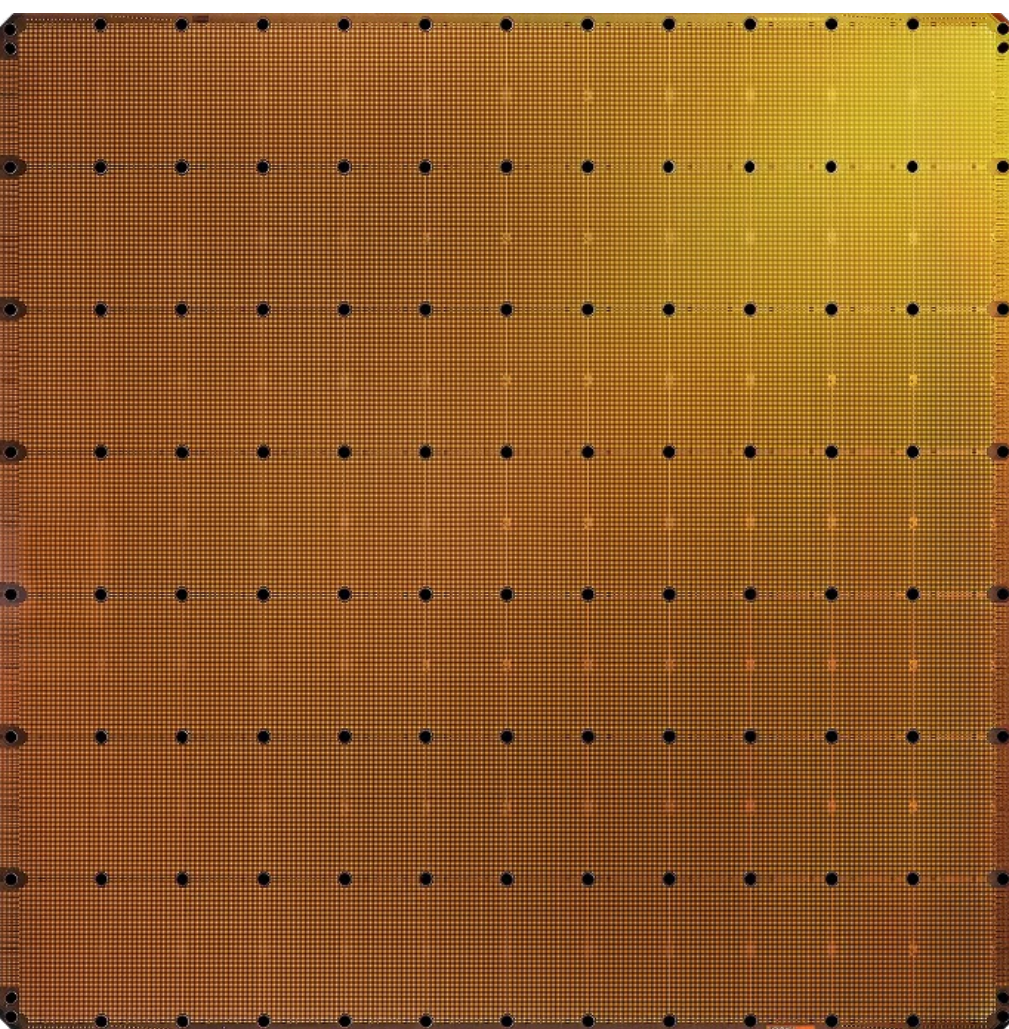


Modern spatial dataflow architectures are designed for AI workloads. Can they be repurposed for general scientific codes?

The Cerebras Wafer-Scale Engine

Built for AI - and the largest chip ever manufactured

- 2D mesh of almost a million processing elements (PEs)
- local per-PE memory (48 KB)
- no global memory



Nvidia B200
for comparison

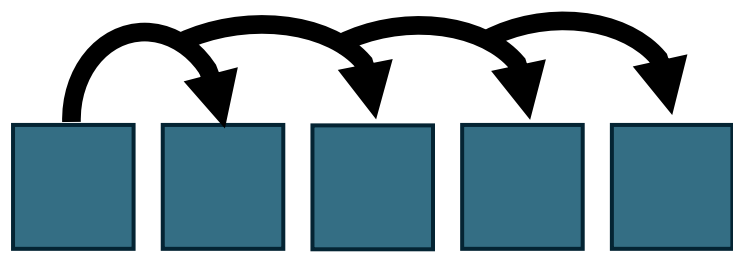
1D Collectives on the WSE

Simple building blocks for

- easy algorithm development
- efficient performance modeling

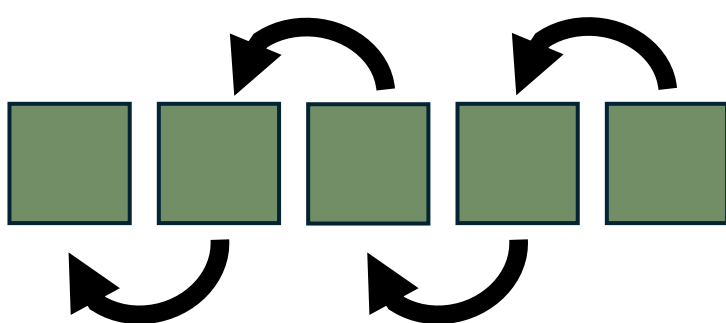
Multicast broadcast:

- at each PE, message is both received and forwarded



Chain reduce:

- compute partial reduction at each hop before forwarding



Sparse matrix-vector product (SpMV) on the WSE

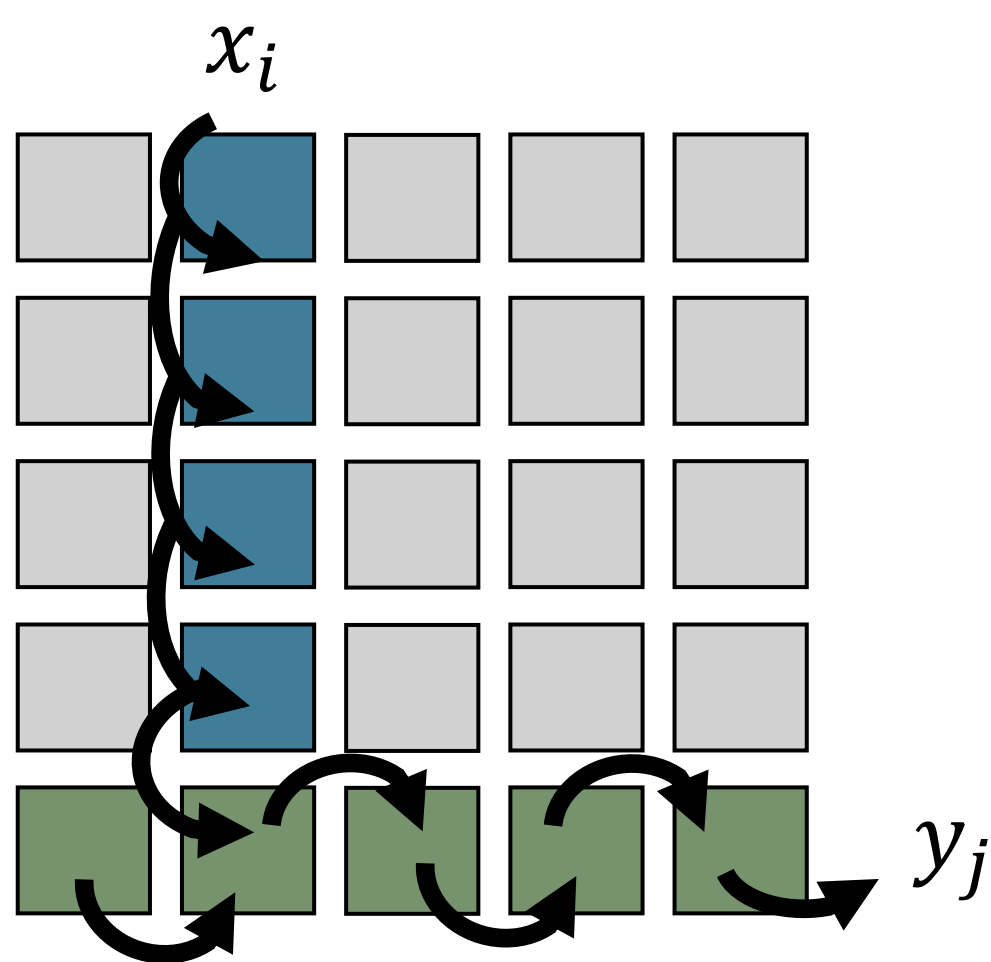
SpMV is a core kernel of many HPC and linear algebra workloads. It also maps well to the 1D collectives, making it a simple and effective research kernel.

Structured sparsity:

- predictable matrix structure can be exploited directly
- enables matrix-free, often stencil-based SpMV implementations
- maps well to spatial dataflow architectures

Unstructured sparsity:

In this work, we develop a general, structure-agnostic algorithm, building on basic 1D broadcast and 1D reduction primitives.



Given a general sparse A,
compute the product:
 $Ax = y$

1. Distribute coefficients of A across the grid
2. Broadcast values of x down the grid columns
3. Compute local partial products at each PE
4. Compute solution values via horizontal reduction

Evaluation in Simulation:

- using the Cerebras cycle-accurate simulator
- at a 64x64 sub-grid size
- at low density, communication closely models full performance
- at higher density, local computation dominates

